

GHOSTPALS

Cyberbrain Neural Interface Qualification

A 7-day work-task gauntlet inspired by Cryptopals and built around Machine Learning
Methods for Neural Data Analysis

YOUR JOB

**Turn raw neural signals into reliable, interpretable cyberbrain
behavior.**

*Functioning systems beat beautiful derivations. But by the end, you must be able to explain
every arrow on the whiteboard.*

TRAINING MANUAL • ML4ND | MODE • HIGH-THROUGHPUT WORKED EXAMPLES | LANGUAGE • PYTHON + PYTORCH

OPERATING PROTOCOL

How we will train

This is not a conventional problem set. The point is to cross the boundary from “I have seen this material” to “I can build with it.” The challenges are sequenced so patterns recur: likelihoods become GLMs, whitening returns in geometry, convolution reappears in deconvolution, and latent variables grow into state-space models.

The Pl2ku protocol

- For most challenges, you do not have to struggle alone first. I can provide a complete, correct implementation so you can run it immediately.
- Then we dissect the code, equations, plots, failure modes, and modeling choices until every component makes sense.
- We deliberately create some broken versions. Failure modes are part of the curriculum, not an embarrassment.
- Later challenges reuse earlier patterns. You should start recognizing recurring “inception patterns” before generating everything from scratch.
- The final challenge of a set becomes a no-peeking task: you implement more of it independently, then we review it together.
- Every day ends with a boyfriend-test / whiteboard explanation: if you cannot explain the model plainly, the ticket is not really closed.

The 7-day arc

DAY	MISSION	CORE IDEAS
Day 1	Probability engine	Likelihoods, priors, latent variables, log-space
Day 2	Recover the neurons	Whitening, detection, clustering, overlapping spikes
Day 3	Signals hidden inside signals	Convolution, deconvolution, calcium factorization
Day 4	Build the neural decoder	Summary statistics, GLMs, Bayes decoding, validation
Day 5	Stop binning time	Poisson processes, conditional intensity, Hawkes models
Day 6	Find hidden brain states	Mixtures, EM, HMMs, temporal latent structure
Day 7	Cyberbrain qualification	Full-stack integration, fault diagnosis, adaptation

DAY 1 // THE PROBABILITY ENGINE

The Probability Engine

INCIDENT REPORT Cortical spike-count telemetry is arriving from an implanted electrode. The inference software runs, but nobody trusts what its numbers mean.

FOCUS Build the probabilistic primitives that everything else will reuse.

CHALLENGE 01 | COUNT THE GHOSTS

Implement a Poisson firing-rate model from raw spike counts.

- Write `log_likelihood(spikes, rate)` and `fit_rate_mle(spikes)`.
- Demonstrate experimentally that the maximum-likelihood rate equals the sample mean.
- Plot the likelihood or log-likelihood as a function of rate and locate its maximum.

Hidden lesson: An estimator is a function of data; likelihood turns a probabilistic model into an objective function.

CHALLENGE 02 | MEMORY FROM BEFORE YOU WERE BORN

Engineering has prior information about plausible firing rates. Add a Gamma prior and watch data gradually overpower it.

- Compute or implement the posterior over the Poisson rate.
- Compare MLE and MAP estimates.
- Repeat for $T = 10, 100$, and $10,000$ observations.

Hidden lesson: A prior matters most when data are scarce; posterior concentration is data winning the argument.

CHALLENGE 03 | TWO MINDS, ONE ELECTRODE

The histogram is bimodal. Assume the system alternates between a low-rate and high-rate latent state.

- Generate a two-component Poisson mixture with hidden state z_t .
- Recover the two rates, mixing probabilities, and state responsibilities/assignments.
- Visualize which observations the model thinks came from which state.

Hidden lesson: Latent-variable modeling is inference about causes you never directly observe.

CHALLENGE 04 | PROBABILITY MELTDOWN

Multiply a huge number of likelihood terms until floating-point arithmetic collapses, then repair the model.

- Create a direct-product likelihood and watch it underflow.
- Rewrite the same computation in log-space.
- Add stable normalization with `logsumexp` where needed.

Hidden lesson: Numerical stability is part of probabilistic modeling, not cleanup.

Certification: Explain why products become sums under logs and why maximizing $\log p$ gives the same optimizer as maximizing p .

NO-PEEKING CHALLENGE

DAY 2 // RECOVER THE NEURONS

Recover the Neurons

INCIDENT REPORT A multichannel cortical implant is streaming extracellular voltage. The operating system needs discrete spike events assigned to biological sources.

FOCUS Move from raw voltage to clean events and neuron identities.

CHALLENGE 05 | CLEAN THE WIRE

Correlated noise makes every downstream distance misleading.

- Band-pass or otherwise preprocess a simulated multichannel voltage trace.
- Estimate the noise covariance.
- Whiten the channels and verify numerically that the covariance is approximately the identity matrix.

Hidden lesson: Z-scoring scales coordinates; whitening also removes cross-coordinate covariance.

CHALLENGE 06 | SOMETHING FIRED

Turn continuous voltage into candidate spike events.

- Implement threshold crossings.
- Keep local extrema rather than every thresholded sample.
- Enforce a refractory period and extract multichannel waveform windows.

Hidden lesson: Event detection is an engineering problem before it becomes a clustering problem.

CHALLENGE 07 | WHOSE SPIKE WAS THAT?

Each event is now a high-dimensional multichannel waveform.

- Flatten each waveform into one feature vector.
- Build or fit neuron templates.
- Compare Euclidean, Mahalanobis, and probabilistic Gaussian assignments.

Hidden lesson: A template is a class-conditional mean; the covariance determines the geometry of “closeness.”

CHALLENGE 08 | TWO NEURONS FIRED AT ONCE

Hard clustering fails when waveforms overlap in time.

- Represent the signal approximately as $X \approx W A$.
- Interpret W as templates and A as time-varying activations/amplitudes.
- Recover overlapping sources and inspect when the factorization succeeds or fails.

Hidden lesson: Source separation introduces continuous latent coefficients instead of one discrete class label per event.

NO-PEEKING CHALLENGE

DAY 3 // SIGNALS HIDDEN INSIDE SIGNALS

Signals Hidden Inside Signals

INCIDENT REPORT Events are not aligned to neat bins. Meanwhile, the optical division sends calcium movies containing overlapping cells.

FOCUS Learn the copy-paste geometry of convolution, then run it backward.

CHALLENGE 09 | THE COPY/PASTE MACHINE

Build convolution from first principles instead of calling a library.

- Create a spike train and an impulse-response waveform.
- Construct the voltage by placing a shifted copy of the waveform at every spike time.
- Verify that the result equals discrete convolution.

Hidden lesson: Convolution is a sum of shifted copies: a structured linear superposition.

CHALLENGE 10 | DECONVOLVE THE GHOST

You are given noisy voltage and known waveform templates. Infer the sparse neural events that produced them.

- Start with one known waveform and a sparse spike train.
- Add multiple neurons, overlaps, amplitude variation, and noise.
- Compare a naive inverse with a sparsity-aware approach.

Hidden lesson: Deconvolution is the inverse problem hiding underneath high-resolution spike sorting.

CHALLENGE 11 | THREE CELLS IN ONE PIXEL

A calcium movie contains spatially overlapping neural sources.

- Represent the movie as $Y \approx F C + B$.
- Recover spatial footprints, temporal calcium traces, and background.
- Visualize each source separately and then reconstruct the movie.

Hidden lesson: Matrix factorization can separate where a source lives from how its activity evolves in time.

CHALLENGE 12 | NEURAL EVENT RECOVERY

Infer spike-like events from a single calcium trace with an exponential decay model.

- Separate observed fluorescence, latent calcium state, latent events, and impulse response.
- Recover events under a plausible decay constant.
- Deliberately use the wrong decay constant and inspect the artifacts.

Hidden lesson: Model mismatch can produce clean-looking but wrong latent events.

NO-PEEKING CHALLENGE

DAY 4 // BUILD THE NEURAL DECODER

Build the Neural Decoder

INCIDENT REPORT You can extract neural events. Management now asks what the activity means and whether it predicts intended behavior.

FOCUS Connect spikes to variables, then reverse the generative arrow.

CHALLENGE 13 | RECONNAISSANCE

Before fitting a big model, interrogate the data directly.

- Implement a PSTH.
- Compute a spike-triggered average.
- Compute a Fano factor and cross-correlation.
- Write one sentence for what each statistic can and cannot tell you.

Hidden lesson: Summary statistics are model diagnostics and hypothesis generators, not obsolete pre-ML rituals.

CHALLENGE 14 | WHAT MAKES THIS NEURON FIRE?

Build a Poisson GLM / LNP-style encoding model.

- Project stimulus features through a linear filter $w^T \phi_t$.
- Pass the projection through a nonlinearity to obtain a nonnegative rate.
- Fit the parameters and recover a receptive field from simulated data.

Hidden lesson: A GLM turns “feature sensitivity” into a probabilistic prediction of spike counts.

CHALLENGE 15 | REVERSE THE ARROW

Cyberbrain control requires inference from spikes back to intended action.

- Simulate neural activity under two possible intended cursor directions.
- Use Bayes' rule to compute $p(\text{direction} | \text{spikes})$.
- Inspect how the posterior changes as evidence accumulates.

Hidden lesson: Encoding asks what spikes should look like given a cause; decoding inverts that model to infer the cause.

CHALLENGE 16 | THE DECODER LIES

A model boasts 93% training accuracy. Your job is to decide whether the celebration is justified.

- Evaluate on held-out data.
- Compare predictive accuracy with neural statistics such as PSTH shape or variability.
- Construct at least one model that predicts labels well but captures the neural process poorly.

Hidden lesson: Prediction and scientific explanation overlap, but they are not the same objective.

Certification: Explain why good classification accuracy does not prove that the generative neural model is good.

NO-PEEKING CHALLENGE

DAY 5 // STOP BINNING TIME

Stop Binning Time

INCIDENT REPORT A command pathway operates on millisecond timescales. Coarse bins are throwing away information.

FOCUS Represent spikes as events in continuous time and model history-dependent intensity.

CHALLENGE 17 | EVENT TIME IS THE DATA

Generate the same homogeneous Poisson process in two ways.

- Simulate with very small Bernoulli/Poisson time bins.
- Simulate directly from exponential inter-spike intervals.
- Compare count statistics and interval distributions as bin width shrinks.

Hidden lesson: A point process is the continuous-time limit of count modeling.

CHALLENGE 18 | A NEURON REMEMBERS ITS LAST SPIKE

Add refractoriness through a history-dependent conditional intensity $\lambda(t | H_t)$.

- Suppress intensity immediately after a spike.
- Let it recover over time.
- Inspect the resulting inter-spike interval distribution.

Hidden lesson: Conditional intensity is a probability model for the next event given everything relevant so far.

CHALLENGE 19 | ONE NEURON WAKES ANOTHER

Simulate a two-neuron Hawkes-style system where neuron A temporarily excites neuron B.

- Define a causal coupling kernel.
- Generate coupled spike trains.
- Estimate or recover the coupling from simulated observations.

Hidden lesson: History kernels turn temporal dependence into an interpretable interaction model.

CHALLENGE 20 | PHANTOM CONNECTIVITY

Now drive both neurons with a hidden common input but remove direct coupling.

- Show that the spike trains become correlated anyway.
- Fit a naive coupling model and observe the false connection it invents.
- Add the common input or appropriate covariates and compare.

Hidden lesson: Correlation can be produced by a third cause; “functional connectivity” is not automatically causal connectivity.

NO-PEEKING CHALLENGE

DAY 6 // FIND THE HIDDEN BRAIN STATES

Find the Hidden Brain States

INCIDENT REPORT Operators report distinct internal modes - rest, preparation, execution, correction - but the neural stream itself is unlabeled.

FOCUS Discover latent structure, then give it temporal memory.

CHALLENGE 21 | NEURAL CONSTELLATIONS

Generate neural population vectors from three hidden network states.

- Fit a Gaussian mixture model.
- Recover means, covariances, mixture weights, and latent state assignments.
- Project the data to 2D or 3D and visualize the clusters and uncertainty.

Hidden lesson: Unsupervised learning compresses high-dimensional activity by positing hidden causes.

CHALLENGE 22 | HARD DECISIONS ARE SUSPICIOUS

Replace hard assignments with posterior responsibilities and implement EM.

- Implement the E-step responsibilities $r_{nk} = p(z_n=k | x_n)$.
- Implement the M-step parameter updates.
- Plot the data log-likelihood after each iteration and verify that it does not decrease.

Hidden lesson: EM alternates between probabilistic inference about latent variables and parameter optimization.

CHALLENGE 23 | THE BRAIN HAS MOMENTUM

Mixtures assume independent latent states. Real neural states tend to persist.

- Introduce a transition matrix $p(z_t | z_{t-1})$.
- Implement the HMM forward algorithm.
- Visualize filtered state probabilities over time.

Hidden lesson: Temporal structure changes the prior over latent states: yesterday's state becomes evidence about today's state.

CHALLENGE 24 | DECODE THE INVISIBLE STATE

You receive only neural observations generated from REST → PREPARE → MOVE → REST transitions.

- Fit or infer the HMM parameters.
- Decode the hidden state sequence.
- Compare inferred states against ground truth and identify ambiguous transitions.

Hidden lesson: State inference is not just clustering with timestamps; the transition model changes the posterior.

NO-PEEKING CHALLENGE

DAY 7 // CYBERBRAIN QUALIFICATION

Cyberbrain Qualification

INCIDENT REPORT The prototype is live. No isolated toy problem remains: the system must survive integration, corruption, and drift.

FOCUS Assemble the full pipeline and prove you can reason about failures.

CHALLENGE 25 | THE FULL STACK

Build an end-to-end toy cyberbrain pipeline.

- Raw voltage → spike detection.
- Spike waveforms → neuron identities.
- Neuron activity → population/latent state.
- Population state → intended movement.
- Expose intermediate outputs so every stage can be diagnosed.

Hidden lesson: The real skill is compositional: previously learned subroutines become modules in a larger system.

CHALLENGE 26 | PRODUCTION FAILURE

I secretly corrupt the pipeline with one or more realistic failure modes.

- Possible faults: correlated electrode noise, overlapping spikes, missing neurons, firing-rate drift, unexpected latent states, or decoder distribution shift.
- Use plots, residuals, statistics, and intermediate representations to localize the fault.
- Repair the correct layer instead of blindly retraining everything.

Hidden lesson: Production ML is often fault localization across a stack, not merely model fitting.

CHALLENGE 27 | THE ADAPTIVE CYBERBRAIN

The biological brain drifts across sessions and yesterday's decoder degrades.

- Compare a frozen decoder, full retraining, and lightweight recalibration.
- Use only a small number of newly labeled calibration trials.
- Measure both recovery and the amount of data/compute required.

Hidden lesson: A useful BCI is a coupled adaptive system: the interface changes while the nervous system changes too.

CHALLENGE 28 | SECTION 9 CLEARANCE

No code at first. The final exam is a whiteboard interrogation.

- Explain: raw signal → latent neural events → probabilistic model → latent state → decoded intention.
- Be prepared for "Why?" at any arrow.
- Then rebuild one complete path in code without copying the earlier solution.

Hidden lesson: If you can explain the structure plainly and rebuild it, the material has become part of your toolkit.

NO-PEEKING CHALLENGE

FIELD REFERENCE

The hidden architecture

probability → signal extraction → convolution → encoding
→ decoding → point processes → latent dynamics

Why this sequence works. Each day deliberately recycles ideas from earlier days. The point is not to remember twenty-eight isolated solutions; it is to feel the same small number of structures appearing in different disguises.

Patterns to notice

PATTERN	WHAT KEEPS REAPPEARING
Likelihoods	A way of scoring model parameters from observed data. They return in GLMs, mixtures, HMMs, point processes, and decoding.
Covariance geometry	Whitening, Mahalanobis distance, Gaussian models, and posterior geometry all ask what directions are stretched, coupled, or uncertain.
Convolution	Shifted copies of an impulse response explain voltage generation and calcium dynamics; deconvolution tries to infer the hidden events.
Latent variables	A mixture assignment, a spike train, a calcium state, and an HMM state are all unobserved variables inferred from measurements.
Bayesian inversion	Forward models describe how hidden causes produce data; decoding and inference reverse that arrow.
Failure modes	Underflow, model mismatch, common-input confounding, overfitting, nonstationarity, and distribution shift are part of the model, not footnotes.

How to start a session

“Pl2ku, Day 1, Challenge 1. I’m at work.”

Then we run the protocol: ticket → synthetic data → complete implementation → run it → inspect outputs → break it → explain it → later re-implement more independently.

Reference material

- Scott Linderman, Machine Learning Methods for Neural Data Analysis: <https://slinderman.github.io/ml4nd/>
- Cryptopals Crypto Challenges: <https://cryptopals.com/>